SIMULATED MICROGRAVITY MODELS SEX DIFFERENCES IN KNEE OSTEOARTHRITIS THROUGH A CD36 REGULATED MECHANOMETABOLIC CIRCUIT

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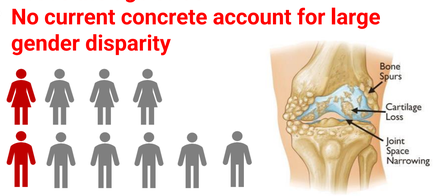
Sex Matters in Knee Osteoarthritis (OA)

- Develops from little to no mechanical loading of the joint.
- The prevalence is **50% of the population above the age of 65**
- **No current concrete account for large gender disparity**

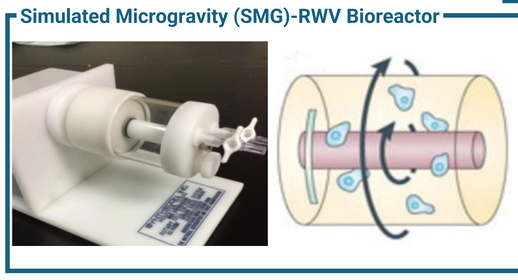


Knee Osteoarthritis (KOA) incident rate and severity is disproportionately higher in females[2]

CD36 and Osteoarthritis

- Prior research suggests that **CD36 may contribute to the inflammatory environment** within the joint.[1]
- CD36-mediated uptake of oxLDL and other ligands can lead to the accumulation of lipid-laden macrophages **(foam cells) in joint tissues.**[3]
- These foam cells release **inflammatory mediators that exacerbate cartilage degradation** and joint inflammation.

Simulated Microgravity (SMG)-RWV Bioreactor



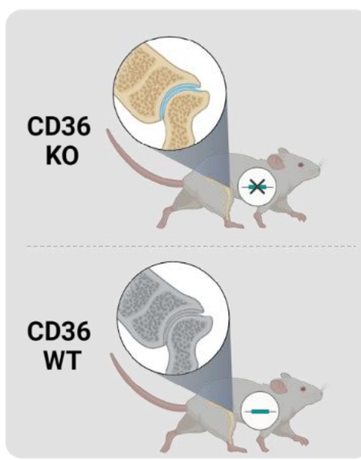
Rotating Wall Vessel (RWV) Bioreactor

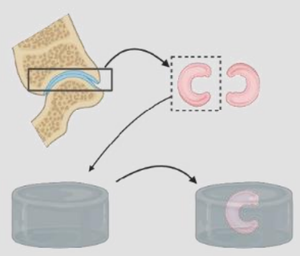
- Horizontally rotating cylinder with a coaxial oxygenator.
- **Induces MG** on the meniscus samples as there is no relative motion of the samples to the media.
- **Novel platform** for replicating a near zero-gravity environment.

Methodology

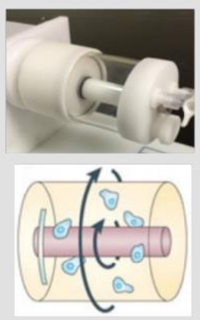
Hypothesis: Long-term exposure to simulated microgravity (SMG) will induce an osteoarthritis (OA)-like phenotype in menisci excised from C57BL/6 mice. Wild-type females are expected to exhibit a more severe phenotype compared to CD36 knockout (KO) females, suggesting a protective role of CD36 deficiency.

Animal Model → Meniscus Encapsulation → Experiment → Analyses

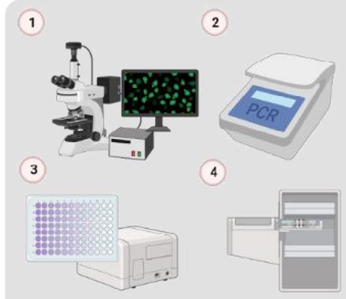




- Meniscus dissection
- Meniscus encapsulation in 1% agarose hydrogel

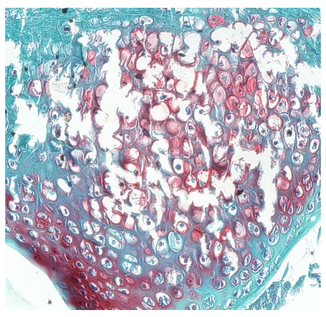


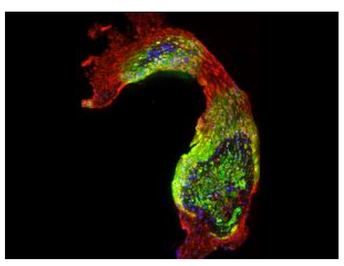
- Simulated microgravity (3 wk)
- Time course Assesment (Wk 0, 1, 2, and 3)



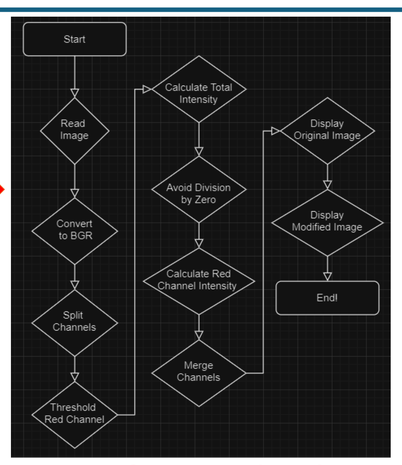
- Histology
- PCR/RNA-Seq
- Biochemistry Assay
- Micro CT

Histology





Collagen I, Collagen II, Nuclei

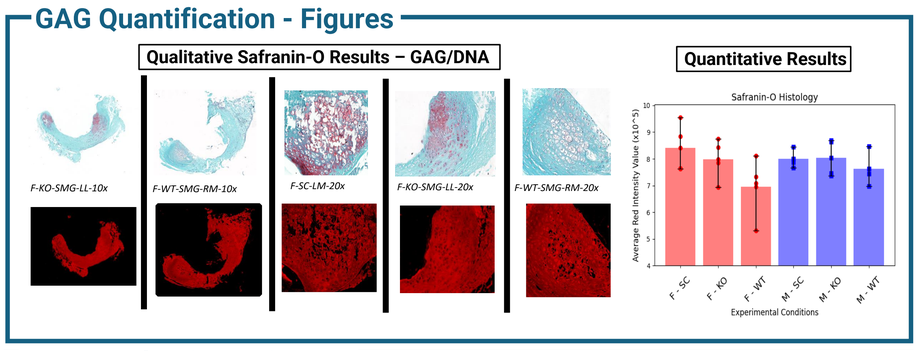


Quantification with Python

Glycosaminoglycan Content (ARIV) is Inversely Proportional to the Severity of Osteoarthritis

Collagen II intensity and distribution in the meniscus is Inversely Proportional to the Severity of Osteoarthritis

GAG Quantification - Figures

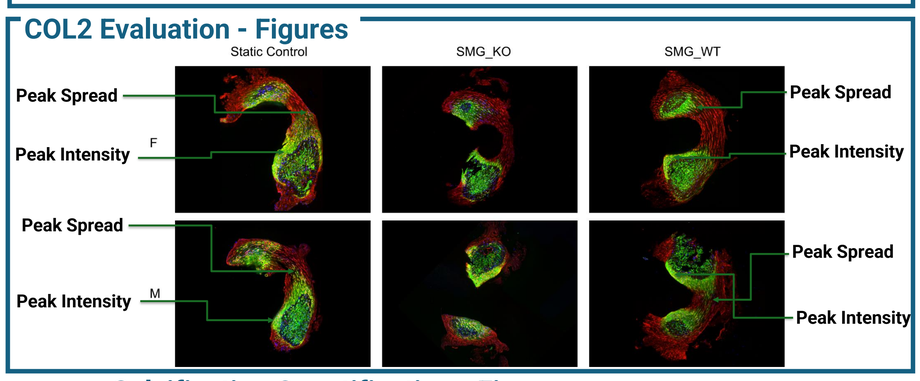


GAG Quantification - Results

- The RVW-SMG Bioreactor **induces a notable decrease in GAG** – namely in females
- The **KO Group in females shows significant increase in GAG content** compared to the SMG group
- Thus, the KO group shows a **protective effect against OA development**

F-SC vs. F-WT	F-WT vs. F-KO	F-KO vs. F-SC
$p < 0.05$	$p < 0.01$	$p > 0.1$

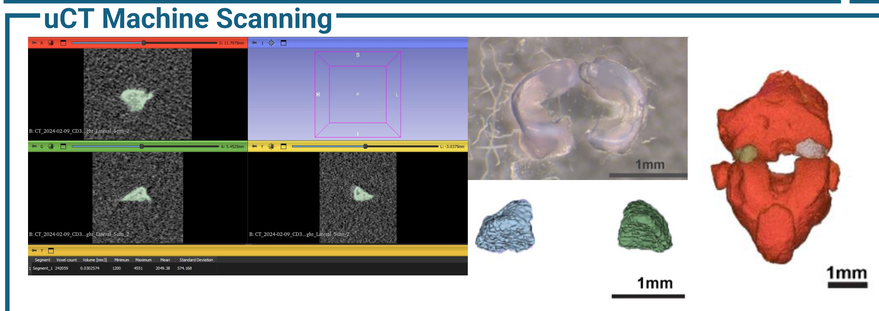
COL2 Evaluation - Figures



COL2 Evaluation - Results

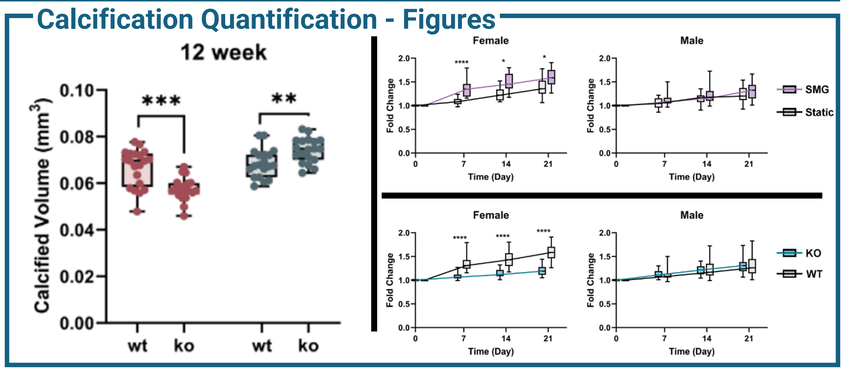
- The RVW-SMG Bioreactor **induces a notable decrease in Collagen II**, especially in females.
- Smaller area and dimmer intensity of Collagen II was observed in the KO and WT group, **alluding to a more aggressive OA phenotype under simulated microgravity.**
- SC group shows more articular cartilage morphology around the periphery; thus, **SC Group is less hypertrophic compared to the SMG group** based on morphology.

uCT Machine Scanning



Calcification Volume is Proportional to the Severity of Osteoarthritis

Calcification Quantification - Figures

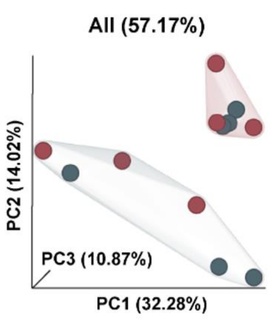


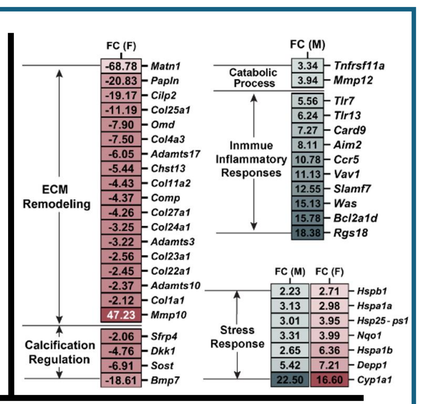
Calcification Quantification - Results

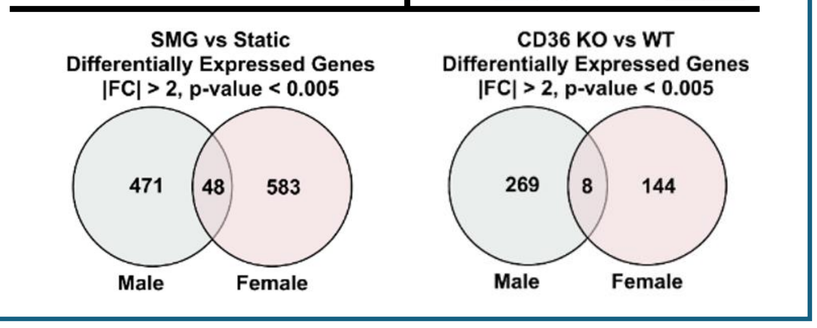
- **Female WT mice** under SMG had **a significant ~25% increase in meniscal calcification volume** by day 7, which remained elevated through days 14 and 21.
- Male WT mice showed less than 10% calcification by day 21.
- **Female KO mice** had **20–30% lower calcification** than WT at all time points under SMG, indicating **CD36 loss is protective**; male KO mice showed no major difference from male WT.

*	**	***	****
$p < 0.05$	$p < 0.005$	$p < 0.0005$	$p < 0.0001$

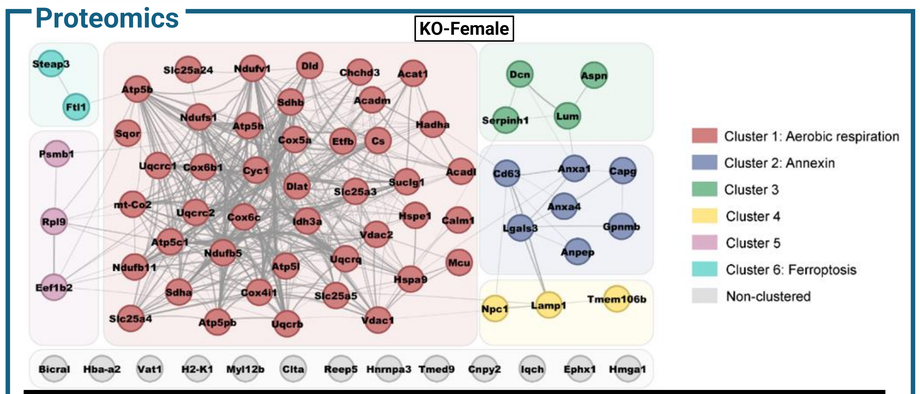
Transcriptomics

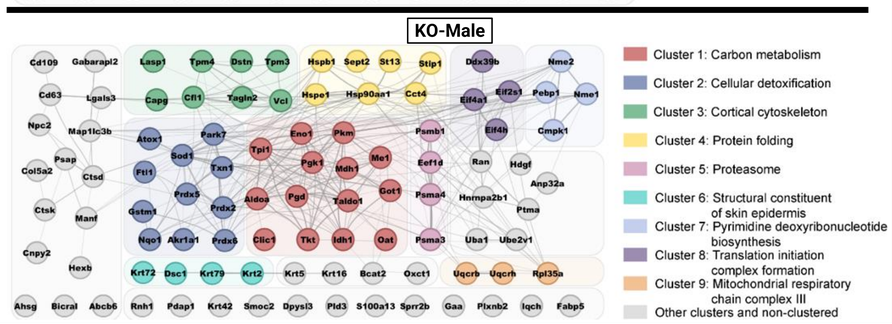






Proteomics



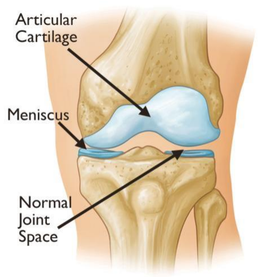


Transcriptomics/Proteomics - Results

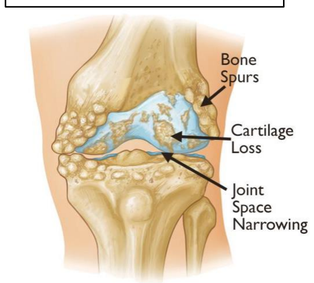
- Principal component analysis (PCA) showed **clear clustering by mechanical condition (SMG vs. static)**
- SMG exposure in females activated potent ECM-catabolic signaling and suppressed structural matrix gene expression, **highlighting a destructive cartilage remodeling response**.
- Venn diagrams at $|FC| > 2$ and $p < 0.005$ indicated **predominantly sex-specific DEGs**, with fewer commonly affected genes in both sexes.
- Female KO menisci showed reduced **mitochondrial activity, fewer inflammatory and matrix-degrading proteins**, and signs of protective metabolic reprogramming under SMG—together **suggesting that loss of CD36 limits tissue damage and calcification in females.**

Final Conclusions

Female KO Meniscus

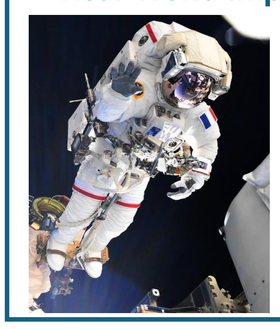


Female WT Meniscus



- The **RWV Bioreactor induces OA-like changes in meniscal tissue under SMG**, shown by reduced GAG and Collagen II and increased calcification—**validating it as a robust in vitro OA model.**
- **CD36 knockout (KO) had a sex-specific protective effect, especially in females**, preserving GAG, reducing calcification, and limiting ECM degradation under SMG.
- Transcriptomic and proteomic analyses showed that **female KO menisci** showed **lower oxidative phosphorylation and fewer catabolic markers**, supporting **CD36 deficiency as protective** against calcification and tissue breakdown.

Real-World Implications



- **Flow Cytometry –** by examining the expression of CD36 in other tissues, **Osteoarthritis can be screened for.**
- **Spaceflight Screening – Astronauts can be screened for spaceflight** with respect to the expression of CD36.
- **CD36 Inhibition –** Due to its proven role in OA development, future drugs could be manufactured **to inhibit the expression of CD36.**

Thank You

Firstly, I'd like to thank **Melanie Kunze, Aillette Mulet Sierra, and Dr. Adesida** for welcoming me to **Adesida Laboratory**, and for teaching me the procedures of the lab. I'd also like to especially thank **Zhiyao (Hilda) Ma** for taking the time to supervise me, and for being there whenever I had questions.

References

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3. Jay, A. G., Chen, A. N., Paz, M. A., Hung, J. P., & Hamilton, J. A. (2015). CD36 Binds Oxidized Low Density Lipoprotein (LDL) in a Mechanism Dependent upon Fatty Acid Binding. Journal of Biological Chemistry, 290(8), 4590–4603. https://doi.org/10.1074/jbc.m114.627026

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